A Progressive Approach to Road Segmentation using Segment Anything Model: From Edge-Aware Enhancement to Fine-Tuned Framework

MAJOR REPORT submitted in partial fulfillment of the requirements

for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
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CERTIFICATE

This is to certify that this project report titled **"A Progressive Approach to Road Segmentation using Segment Anything Model: From Edge-Aware Enhancement to Fine-Tuned Framework"** is a bonafide record of work done by **GAYATHRI VINUKOLLU (228W1A12J4), RAJASEKHAR SOLA (228W1A12J2)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	A Progressive Approach to Road Segmentation using Segment Anything Model: From Edge-Aware Enhancement to Fine-Tuned Framework
2	Student Names & Numbers	Ms. Gayathri Vinukollu 228W1A12J4 Mr. Rajasekhar Sola 228W1A12J2
3	Name of The Guide	Dr.Ch. Subba Reddy
4	Research Group	CV/IP
5	Application Area	Navigation Systems, Urban Planning, and Autonomous Vehicles
6	Aim of the Project	To enhance road network extraction by using the Segment Anything Model (SAM)
7	Project Outcomes	Implemented an optimized SAM-based framework that enhances road segmentation precision, connectivity, and reliability across diverse satellite imagery environments.

Student Signatures

1.

2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr.M. Suneetha**, Dean R&D, Professor & Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr.Ch.Subba Reddy, Assistant Professor** for **his** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficult on my part to finish the project work.

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ABSTRACT

Satellite imagery is often used for extracting roads to facilitate tasks such as urban planning, navigation, and disaster management. However, traditional segmentation methods do not work well in certain areas because roads vary in width, are occluded by other objects, and there is a lot of complexity within urban environments. SAM has a high level of generalization; however, its performance in segmenting roads is limited when it does not receive any task-specific prompting. This paper develops progressively on models based on SAM, beginning with a baseline SAM model and prompting it using edge-aware prompts, then hybridizing a DeepLabV3 model and SAM to improve semantic understanding. Using the information gathered from the previous steps, a fine-tuned model based on SAM is presented to further improve segmentation accuracy. Ultimately, the proposed methodology provides a means of accurately segmenting both wide and narrow roads, increasing connectivity among road segments, and reducing the number of false positives in both rural, semi-rural, and urban dense environments. Results show that the fine-tuned model outperformed both the baseline and hybrid models with regard to accuracy and structural consistency. To summarize, results demonstrate that adapting the prompting type and method utilized with SAM provides a significant advancement in road extraction tasks when utilizing a general-purpose model for this purpose.

Keywords

Road Segmentation, Satellite Imagery, Segment Anything Model (SAM), DeepLabV3, Remote Sensing, Fine-Tuning, Edge-Aware Filtering, Computer Vision

CHAPTER – 1

Introduction

1.1 Origin of the Problem

The development and growth of deep learning approaches have greatly helped to achieve better results in the process of image segmentation, especially in road extraction from satellite images. Hybrid approaches like DeepLabV3-SAM [1], where semantic segmentation is combined with prompt-based refinement, contribute to achieving better results through taking advantage of the strengths of both methods. The Segment Anything Model (SAM) has also been used to extract road networks in graphs [2].

Convolutional neural network approaches still remain widely used when working with road extraction. There are more effective approaches to the DeepLabV3+ model [3], which utilize the advantages of multi-scale context aggregation during feature extraction. Other effective approaches are optimized versions of CNN-based segmentation models [4], and also there exist some modifications of the Segment Anything Model [5].

Graph-based methods emphasize the preservation of road connectivity and topology. Such approaches as road graph extraction using Graph Neural Network [6] and topology representation using transformer architecture [7] aim at ensuring consistent topology of roads. Moreover, advanced solutions, like RNGDet++ [8], improve road network topology through multi-scale feature learning. Finally, one-stage solutions [9] simplify the road extraction by estimating both the geometry and the connectivity of the roads simultaneously.

Prior deep learning models, such as Deep Residual U-Net [10] and D-LinkNet [11], helped improve the segmentation performance through effective feature reuse and encoding. Self-supervised algorithms [12] provide for better generalization of the models by reducing the dependency on labeled datasets.

Methods focusing on topology, such as RoadTracer [13] and DeepRoadMapper [14], introduce graph representation of road networks in contrast to pixel-based approach. As a

consequence, benchmarks such as DeepGlobe [15] become necessary for the evaluation of road extraction performance.

Transformer-based segmentation networks including SegFormer [16], Mask2Former [17] and others perform better because they leverage global context information. Modern solutions including SatMAE [18] and transformer pre-training for multispectral imagery [19] provide better generalization capability with self-supervised learning technique.

Moreover, mask-based and transformer approaches including RemainNet [20] and DOCswin-Trans [21] can be used to improve results of segmentation tasks. SpaceNet dataset can also be used to test the effectiveness of various algorithms.

However, despite the described advantages of modern approaches, road segmentation is still an open problem that requires special attention. It can be explained by several factors. First, roads are elongated objects of different widths and textures. Moreover, occlusions caused by vegetation, buildings, cars, lighting effects, and others increase difficulty of segmentation process.

To solve the described problem, a progressive framework for road segmentation is suggested in the project. The proposed methodology involves baseline SAM testing, edge aware approach enhancement, DeepLabV3-SAM algorithm implementation and final fine tuning of the model.

1.2 Basic Definitions and Background

A basic knowledge of important terms pertaining to satellite images, how to segregate images by segment, and how models based on deep learning perform extraction tasks will enable you to comprehend the methodology that follows. These three areas provide the basis of your understanding when attempting to read further chapters.

Satellite Imagery

The use of remote sensing satellites for imaging the earth's surface through either multispectral or optical sensors is referred to as satellite imagery. These images are used in multiple applications, including land-use studies, environmental monitoring, urban

development, and mapping infrastructure and provide spatial and contextual information about geographic areas.

Despite their usefulness, automated analysis of satellite imagery presents many difficulties as follows:

- Significant differences in resolution and quality between various images;
- Noise, shadows, and distortion will always affect an object's detected location;
- Partial or complete obscuration by both natural and artificial objects (occlusions) creates difficulty;
- Complex backgrounds that share visually similar attributes.

Consequently, feature extraction, specifically road detection from satellite imagery, is a very challenging process because of all the above factors.

Road Extraction

The goal of road extraction is to identify roads from satellite imagery (or other remote sensing imagery) and produce structured outputs (e.g. binary masks, graph-based networks) that represent them. The goal is to detect road pixels correctly and to ensure connectivity, continuity, and geometric shape integrity when doing so.

A successful road extraction system must have the ability to:

- Detect both wide and narrow roadways
- Have continuity between roads that intersect or have curves
- Be able to process occlusions and low contrast areas
- Reduce the number of mis-detected objects due to their similarity to roads.

Image Segmentation

In image segmentation, pixels of information in the image are classified and grouped into meaningful sections of the overall image. In the following, segmentation will be used to

identify which pixel belongs to the roadway and which pixel does not belong to the roadway.

There are two major approaches for performing segmentation based on the previous description:

Semantic Segmentation:

This type of segmentation assigns class labels to every pixel within an image. There are several models that perform semantic segmentation, such as DeepLabV3+, which use convolutional neural networks combined with atrous convolution and multi-scale feature extraction methods to capture the contextual relevance of areas within an image. These models are very good at detecting and identifying large structures within an image; however, the outputs can be coarse and often lack precise-edge precision.

Prompt-Based Segmentation:

Prompt-based segmentation provides an example of an interactive approach to performing segmentation. Inputs such as points, bounding boxes, and/or masks give guidance for performing the segmentation. This approach allows for tailored segmentation to higher levels of accuracy and resolution, particularly in settings where the conditions are complex and automated methods do not provide satisfactory results.

Segment Anything Model (SAM)

The Segment Anything Model (SAM) is a general-purpose model designed for the image segmentation task. The model is constructed out of three main components which will be discussed in more detail below:

- An image encoder (built on top of a vision transformer).
- A prompt encoder (to process the user's input).
- A mask decoder (to produce the segmentation output).

Utilizing this design, SAM is capable of generating hundreds of segmentation masks from images produced in various situations. It offers prompt-driven capabilities, giving users the option to limit their focus to specific areas of interest.

Within this work, SAM is responsible for both the refinement and generation of the final segmentation. Additionally, after incorporating domain-specific training via fine-tuning, SAM will become able to recognize road structures through its own identification process while providing more accurate and reliable results.

Semantic Segmentation Models (DeepLabV3)

The DeepLabV3 semantic segmentation model has become one of the dominant models of its kind by virtue of its use of atrous convolution and ASPP to provide a multi-scale context. It is effective for the segmentation of large objects, and it provides an effective baseline for determining road area.

However, this model has some serious limitations:

- It struggles with detecting thin & narrow objects
- Complex and occluded areas result in fragmentation
- Poor boundary refinement.

The deficiencies outlined above suggest that there is an opportunity for both additional refinement approaches and methodologies to be pursued.

Image Preprocessing Techniques

The purpose of an image preprocessing process is to achieve an improved set of input data to use in conjunction with the computer vision task that you plan to use the images for. The following techniques are commonly used to obtain improved road attributes from substandard input [i.e., low contrast, noisy input]: contrast enhancement, high pass filtering, and edge detection. These techniques allow for improved feature extraction, improved prompts for image generation, and improved segmentation accuracy.

Fine-Tuning of Foundation Models

Fine-tuning refers to the method of modifying a previously trained model to fit a specific dataset, in this instance, the Segment Anything Model is being fine-tuned using satellite imagery datasets to better recognize road infrastructure.

The following tasks are completed as part of the fine-tuning process:

- Adaptive prompts are generated from ground truth masks;
- Selected portions of the model are trained;
- Segmentation output is optimized with the use of loss functions and evaluation metrics.

Through fine-tuning, the model is able to:

- Learn domain-specific attributes
- Have improved detection of thin and occluded roads
- Reduce reliance on third-party segmentation models.

Summary of Conceptual Flow

A conceptual progression of the study is as follows:

- Improvement of input images in order to provide better visibility of features
- Utilization of segmentation models to extract coarse road structures
- Use of prompt-based approaches to improve the quality of segmentation
- Development of a unified approach via fine-tuning of a foundational model.

Through each step in the conceptual progression, there exists a systematic effort to increase the quality of segmentation and reduce the complexity of the segmentation process while providing a viable and efficient solution for road extraction.

1.3 Problem Statement

It is challenging and complex to extract information about roads from high-resolution satellite images because they can be seen under many different real-world conditions and they can be modified often. Roads are usually long, narrow, and continuous, and their width, texture, and orientation can be very different from one another. The appearance of roads at times is so similar to adjacent features that have an open area of bare ground or

vacant land next door, as well as with buildings, roofs, and parking lots; therefore, automated systems must be able to distinguish between the two types of features accurately.

The major difficulty with road segmentation is occlusion. Roads are often occluded by trees, vehicles, shadows, or other structures, which can lead to broken sections of the road and poor results for segmentation. Road networks can be broken due to occlusions because a road may not continue to be visible due to an occlusion. In addition to occlusions, differences in the amount of light, weather conditions, and quality of the satellite sensor can create noise and inconsistency in satellite images, which will further reduce the performance of the segmentation.

The problem with the traditional method of image processing is that it uses established characteristics based on predetermined rules and thresholds, which are ineffective for dealing with many of these issues. The performance of deep learning-based semantic segmentation models has improved considerably as a result of hierarchical representations. However, they are unable to capture fine detail's essential characteristics. For example, these models produce outputs that typically do not completely capture the thinness or rugged quality of the road surfaces and hence will produce inadequate maps of the road network if the road network generated using a deep learning-based model has little or no completeness or continuity.

Alternatively, a newly developed segmentation technique, called "prompt-based segmentation," enhances flexibility and control over the direction provided to segmentation models. At this time, though, prompt-based segmentation models can be adapted to specific regions; nevertheless, large amounts of assistance from sources outside the model are required for identifying road features when using satellite imagery. Likewise, prompt-based segmentation is very dependent upon how well the prompts and any necessary preprocessing procedures are developed, and therefore, the inconsistency of the prompts and preprocessing procedures will limit the reliability of prompt-based segmentation systems.

Specialized multi-phased approaches using multiple models yield better quality segmentation at the expense of increased computation, a longer inference time, and a

reliance on dozens of different components in the system. Such systems also do not scale well and often do not meet the requirements of real-time or large-scale applications.

The key issue addressed by the work reported here is:

a. Develop an accurate, robust, and efficient method for extracting roads from satellite images that extracts continuous, accurate, and consistent road segmentation while reducing the complexity of the model as well as reducing the use of multiple preprocessing stages.

To meet this problem, the proposed work will focus on fine-tuning a basic model through domain-specific adaptation. Specifically, the model will:

- Correctly identify both large and small road features;
- Maintain connectivity through occluded and complex areas;
- Increase the precision of the road boundary and the quality of the segmentation;
- Decrease reliance on existing external models and multi-staged pipelines; and
- Provide a scalable, efficient solution for real-world applications.

The objective of the proposed approach is to address the limitations of the major road segmentation models currently available and provide a cohesive framework for high-quality extraction of the road network.

1.4 Applications

Satellite images can provide accurate road locations, with a large number of beneficial uses in the real world. By automatically identifying and mapping out road systems, it allows for more efficient planning, monitoring, and decision-making across several industries. Having improvements to the accuracy and continuity of the proposed approach improves the reliability of these types of projects.

Urban and Regional Planning

To effectively build and maintain infrastructure, urban planners are reliant on accurate data about road networks. Road extraction via automation aids in the following areas:

- Monitoring urban growth and improvements to infrastructure

- Planning for future road development and transportation networks
- Recognizing connectivity issues in the current road system

The increased segmentation quality allows for the accurate capturing of smaller and/or newer roads showing that they are more useful when making planning decisions.

Disaster Management and Emergency Response

Roadways are a critical factor when providing rescue and relief efforts after disasters such as floods, earthquakes, or mudslides. The fast and accurate identification of roadways can help with identifying:

- Roadways that can be accessed by emergency vehicles
- Roadways that are damaged or blocked
- Efficient planning of evacuations and rescues

Having an effective segmentation system allows for reliable roadway identification even when environmental factors make it difficult to do so, e.g., debris, water, or partial obstructions on the road.

Navigation and Autonomous Systems

To operate properly, automated navigation systems depend heavily on accurate road information. Therefore, automated road extraction provides the following benefits:

- Updating digital maps with current road information
- Increasing the accuracy of navigation and route planning
- Providing accurate road information to help autonomous vehicles navigate safely and efficiently

In order to provide safe and efficient navigation and routing, both fine detail and continuity of road information are required at each step throughout the process of creating digital maps.

Geographic Information Systems (GIS)

Spatial Data: Maintaining General Organization Includes Like Roads

GIS provides analytical and visual tools for precise spatial data. Road extracted is for:

- Spatial Analysis and Mapping
- Organizing Infrastructure
- Transportation / Logistics

Segments of high quality (initially) will also increase the accuracy of GIS models and analytical methods.

Smart Cities and Infrastructure Monitoring

Real-time data is necessary for smart city initiatives in managing urban infrastructure. Automation for extracting road networks from imagery supplies valuable data resources to support:

- Motor vehicle traffic monitoring and optimization;
- Infrastructural maintenance and planning;
- Integrating with IoT-enabled smart systems — all of which improve decision-making effectiveness while enhancing resource allocation for optimal utilisation.

Dataset Annotation and Model Training

Road datasets have traditionally required a lot of manual effort to create accurate, usable records due to the time-consuming and labour-intensive nature of annotating them. However, with the implementation of automated techniques to extract road information from aerial imagery, automated methods provide:

- High-quality masks for training deep learning networks;
- Reduced costs and effort associated with producing datasets
- Faster development of new computer vision applications.

As a result, the new approach proposed here can help create annotated datasets for use by various organisations in the future.

Defence and Surveillance Applications

In essential defence and surveillance planning, road networks are key sources of information. They are also used by both the military and civilian governments for future planning. Road network data can be applied in several ways, including:

- analysis of routes used to conduct military operations;
- surveillance and monitoring of remote or sensitive infrastructure.
- conducting reconnaissance/surveillance operations.

Gain access to road feature information on an accurate and timely basis will enhance operational performance as well as situational awareness.

Summary of Application Impact

There are several benefits associated with the proposed framework for road extraction including high accuracy, great reliability, increased ability to detect complex/thin roads, scalability, and efficiency. These benefits allow for the implementation of this approach into actual systems where an accurate automated road network can be produced.

CHAPTER-2

Review of Literature

2.1 Description of Existing Systems

[1] DeepLabV3-SAM: A Novel Image Segmentation Method for Rail Transportation

Authors: Li Wei, Haruto Nakamura, Dr. Elisa Romano

Published Year:2023

Observations:

Li Wei, Haruto Nakamura, Dr. Elisa Romano DeepLabV3-SAM: A Novel Image Segmentation Method for Rail Transportation [1] propose a new hybrid segmentation framework referred to as DeepLabV3-SAM that utilizes the strengths of DeepLabV3 for semantic segmentation to generate coarse semantic maps of railways, structures, and adjacent modes of transportation. Using atrous convolution, DeepLabV3 is able to produce multi-scale representations of the aforementioned classes. The Segment Anything Model (SAM) takes the coarse masks generated by DeepLabV3 and produces more accurate object boundaries for the given masks based on user-supplied prompts such as points and/or bounding boxes, thus improving the accuracy and definition of identified objects [1].

DeepLabV3-SAM was tested on the RailSem19 dataset and outperformed both DeepLabV3 and U-Net in all metrics. In particular, the IoU score increased up to 10% on the large rail structures that had clearly defined structural boundaries when using SAM to refine the objects [1]. Only minor dataset-specific retraining is needed with this model due to its architecture, and therefore it can provide reliable results even with a limited amount of training data. The system will provide high precision results, reduced labeling costs, and is adaptable to varying light and occlusions depending on the location of the object. Overall, this system will be beneficial for automated rail monitoring, automated rail systems, and automated rail infrastructure inspection applications [1].

[2] Road Extraction Based on Improved DeepLabV3+ in Remote Sensing Images

Published Year: 2022

Observations:

This research Road Extraction Based on Improved DeepLabV3+ in Remote Sensing Images [2] proposes a modified version of DeepLabV3+ for extracting roads from high-resolution remote sensing data. The modification includes using a ResNet backbone in conjunction with an improved ASPP module to improve the extraction of contextual and multiscale features from the imagery [2]. These improvements aid in identifying roads that are very thin and/or are discontinuous or occluded and were often missed by traditional methods.

Extensive testing on a number of satellite imagery datasets shows that the revised version of the model produces better continuity, accuracy, and overall IoU results than standard DeepLabV3+ and U-Net models [2]. The improved feature aggregation mechanism of this model allows it to maintain the continuity of roads even in difficult environments (e.g., covered by vegetation, buildings casting shadows, complex urban forms, etc.). Even with the performance improvements, there are still shortcomings in the model's ability to detect extreme narrow roads and heavily occluded areas. However, the model will likely be of significant interest to GIS professionals, urban planners, transportation mapping practitioners, and researchers working in automated cartography due to its overall stability, enhanced feature representation, and high level of generalization [2].

[3] SAM-Road: Segment Anything Model for Road Network Graph Extraction

Authors: Ananya Mishra, Leo Zhang, Fatemeh Sadat Saleh, Dr. Erik Olson

Published Year:2024

Observations:

A new adaptation of the Segment Anything Model (SAM) to generate vectorized road networks from satellite images was developed by SAM-Road [3]. The approach deviates from the traditional methods of segmentation and graph creation as this approach

repurposes SAM's very powerful image encoder mode for semantic segmentation to identify roads & intersections accurately. Non-maximum suppression is used to process the segmentation results into graph vertices efficiently. A small transformer-based GNN is used to perform the topology prediction using rich feature embeddings from the SAM image encoder to help assess edge connectivity [3]. The most significant feature of SAM-Road is its simple yet powerful implementation without complicated post-processing and/or graph building in multiple stages. The improvement in speed is substantial as the model has achieved a performance improvement factor of 40 times above RNGDet++ on the City scale dataset and achieved similar accuracy. The model can create complete road networks across several kilometers2 within seconds and can be utilized for real time applications such as autonomous navigation and smart city planning [3]. This study demonstrates the possibility of foundational vision models like SAM being beneficial when they are utilized for graph-based learning tasks. SAM-Road sets new standards with regards to performance and simplicity and that foundational models can be successfully expanded for structured data prediction [3].

[4] Road-SAM: A Universal Segment Anything Model for Road Extraction from High-Resolution Remote Sensing Images

Authors: Dr. Xiaoyu Feng, Rong Li, Harshita Rao, Dr. Chen Yang

Published Year: 2024

Observations:

Road-SAM [4] provides a flexible and scalable means of extracting roads (road extraction) from very high resolution (VHR) optical remote sensing images while also altering how generic vision models (e.g., Segment Anything Model, or SAM) can be modified for geo-spatial purposes. Traditional road extraction techniques depend on custom architecture and/or large scale domain-specific fine-tuning; Road-SAM uses three customizable adapter variations embedded in the transformer blocks of SAM to adapt to Road-SAM. The locations of adapters within the model are strategically chosen to incorporate specific features to be refined from the model, allowing for increased flexibility and precision in how the model processes data. In addition, similar to other road extraction models that

utilize visual prompting, the key innovation of Road-SAM is the use of Explicit Visual Prompting (EVP) and the embedded features and high frequency component (HFC) data used in training are being utilized as visual prompts, allowing for more robust guidance in directing the model's focus toward road-like structures [4]. Road-SAM's frequency adapter fine-tuning mechanism updates only 10% of the network model parameters, making it a lightweight and computationally-efficient alternative to current state-of-the-art methods that achieve the same level of accuracy [4]. Road-SAM was superior to current state-of-the-art methods with respect to both accuracy and adaptability on the benchmark dataset tested. Road-SAM with its extensive ablative studies shows how each modular component contributes to the overall architecture, making it not only a powerful segmentation tool but also cost-effective due to its modularity and adaptable design; thus, making Road-SAM a valuable tool for remote sensing applications including smart urban mapping and automated GIS updates [4].

[5] Enhancing Road Scene Segmentation with an Optimized DeepLabV3+

Authors: Raghav Sharma, Nisha Mohan, Dr. Alejandro Ruiz
Published Year: 2024

Observations:

The authors Raghav Sharma, Nisha Mohan, Dr. Alejandro Ruiz of enhancing Road Scene Segmentation with an Optimized DeepLabV3+ [5] have developed a lighter, more efficient, optimized version of DeepLabV3+ with the goal being to improve segmentation ability to perform real-time analysis of the road from a camera. The new model can utilize a MobileNetV2 backbone rather than the more expensive Xception backbone; thus, this change will reduce the overall complexity of the model, yet still produce very strong features. In addition, to continue to enhance performance of the new model, the authors redesigned their ASPP module in order to obtain better contextual features with different receptive fields, as well as integrative functions of the Shuffle Attention module to allow for small to fine features to become more sensitive [5].

Overall, the optimized model produces superior performance of thin features (e.g., lane boundaries, road markings, traffic poles, and edge features) and has historically been

difficult to detect using traditional segmentation networks than under otherwise reasonable conditions. Furthermore, the optimized model is ideal for Real-Time Autonomous Driving applications, Smart Traffic Management, and Onboard Road Analysis due to the speed of inference. Conversely, the optimized model currently has limitations if exposed to dense crowd scenes and/or extreme shading and severely lacks visibility due to low lighting [5]. Ultimately, this research will allow the development of practical-size deployment solutions to balance accuracy and efficiency in the next generation of Intelligent Transportation Systems [5].

[6] End-To-End Road Graph Extraction Based on Graph Neural Network

Authors: Chengkai Yang; Ion-George Todoran; Christian Saravia

Published Year: 2023

Observations:

RoadGraphNet project aims to create new pathways for creating a road map from satellite images, a highly challenging task because of occlusions and highly complex urban layout, as well as being complex to use more traditional methods of extraction - which utilize traditional rule-based segmentation followed by a rules-based graph creation. RoadGraphNet's solution provides an end-to-end method of extracting a road graph utilizing deep learning; this includes utilizing Convolutional Neural Networks (CNN) to find the centreline key points (i.e., points of interest) of the road such that the RoadGraphNet can construct the road graph using Graph Neural Networks (GNN). Thus, RoadGraphNet is able to build the road graphs without requiring complex post-processes which can lead to inaccuracies in connecting and constructing a more topologically consistent road graph [6]. RoadGraphNet also provides an efficient method for processing images at the city level, while the algorithm operates with high performance; thus, providing robust solutions for the large variability of road conditions such as intersections, occlusions, and variations in brightness from both natural light and artificial light sources. The model outperformed previous methods in precision, recall, and F1-score by being trained on multiple high-resolution aerial images collected from around the United States, for use within determining graph connectivity/completeness of street networks in

Washington, D.C., New York, San Francisco; along with visual results of new and improved road networks [6]. The model predicts street graph data via an efficient single pass methodology that streamlines existing processes for larger implementation scale of non-traditional urban infrastructure projects such as autonomous vehicles. This provides a practical & transferable approach, ensuring all raw data is converted to usable vector-based street network map representation instantly [6].

[7] Road Graph Extraction via Transformer and Topological Representation

Authors: Yifan Zao; Zhengxia Zou; Zhenwei Shi

Published Year:2023

Observations:

Yifan Zao; Zhengxia Zou; Zhenwei Shi, proposed Road Graph Extraction via Transformer and Topological Representation that is Road2Graph[7] offers a revolutionary method to extract road networks from remote sensing images by directly encoding and predicting the topology of roads in an end-to-end process. Legacy methods for building road networks have historically faced challenges with continuity and efficiency when working with extremely complex landscapes such as urban areas. With the use of transformer-based architectures, Road2Graph combines deep convolutional features and multi-scale information in order to create a comprehensive understanding of topological relationships in large scale road networks [7]. Additionally, the architecture creates a compact, expressive representation of road topology that captures important characteristics like road segmentations, vertex positions, midpoints, and interconnectivity, all while minimizing the redundancy of parameters. This representation can be rapidly decoded into a road graph, promoting high-speed, high-fidelity extraction. When conducting experiments with CityScale and SpaceNet data sets, Road2Graph has demonstrated superior performance over previous state-of-the-art methods, yielding unprecedented gains in accuracy (TOPO-F1: +1.55% and +2.23%) and continuity (APLS: +7.03% and +3.05%) [7]. Ultimately, the success of Road2Graph exemplifies the transformative power of transformer models on understanding spatial structure and predicting topology of geospatial analysis, and as such, is well-suited for smart city development, autonomous systems, and GIS [7].

[8] RNGDet++: Enhanced Road Network Graph Detection with Multi-Scale Supervised Learning

Authors: Zhenhua Xu; Yuxuan Liu; Yuxiang Sun; Ming Liu; Lujia Wang

Published Year:2024

Observations:

Zhenhua Xu; Yuxuan Liu; Yuxiang Sun; Ming Liu; Lujia Wang proposed RNGDet++: Enhanced Road Network Graph Detection with Multi-Scale Supervised Learning [8] and it is been developed as an enhancement to automatic road network graph detection, focusing on overcoming the drawbacks from manual annotation, and deficiencies of prior graph based algorithms . One of the primary reasons for developing RNGDet++ is that road network graphs are very important for autonomous vehicle systems; in particular, for tasks associated with navigation and global route planning (e.g., finding the shortest path from point “A” to point “B”). With this knowledge, RNGDet++ enhances the original model by adding an instance segmentation head which improves the model's supervision during training while providing better modeling of the multi-scale characteristics of the road's structural components via backbone feature extraction [8]. Historically, most previous approaches to the detection of road networks have relied on either post-processing semantic segmentation outputs and/or rule-based graph predictions, both of which have created rules that contain a lot of hard-coded logic with suboptimal results based on performance. RNGDet improves over existing methods by combining information learned directly from both the segmented surface of the roads and the road structure represented in graph form to create more accurate and complete overall road path mappings of the data collected. Using two separate publicly available benchmark datasets, RNGDet has outperformed all competing approaches on nearly every evaluation provided by the two datasets demonstrating the capability of using multi-scale feature fusion combined with supervisions at the instance level [8]. Furthermore, RNGDet's improved performance will not only allow for improved map production, but will also provide the basis for enabling real-time automated road network generation, which are both essential components of autonomous driving systems. Through providing a substantial body of empirical evidence

along with an efficient architecture, RNGDet sets the new standard for scalable, accurate road network extractions [8].

[9] Single-Shot End-to-end Road Graph Extraction

Year :2024

Authors: Gaetan Bahl , Mehdi Bahri ,Florent Lafarge

Observations

Gaetan Bahl , Mehdi Bahri ,Florent Lafarge,In this work Single-Shot End-to-end Road Graph Extraction [9] a novel one-shot learning approach is suggested to construct a graph of roads based on the information derived from satellite images. Contrary to other works where there were several steps to build a graph, the suggested one enables building such a graph without iterations because we make predictions concerning the key points of roads and their interconnection in just one step. The main idea of the suggested one-shot learning approach is related to applying CNN for building a graph based on the data extracted from satellite images, which will allow us to benefit from the spatial relationships of road segments [9]. According to the results of experiments performed, we may assume that our model is highly efficient because the time taken to conduct predictions is relatively small compared to other models. However, this approach is not less effective than other methods despite its increased speed. However, due to the effectiveness of this method, the architecture experiences issues with identifying the right topology in the case of high-density environments that have multiple junctions or intersections because of the inability of the architecture to differentiate between multiple junctions. Additionally, this method lacks the ability to generalize its knowledge and apply it to longer roads. Thus, the use of this method will be problematic when there is occlusion on the road [9]. In spite of all the above limitations, it can be argued that this architecture provides a robust and effective way of forming a road graph [9].

2.2 Summary of Literature Study

Table 2.1: Summary of Literature Study

Paper Title	**year of publication**	**review**
DeepLabV3-SAM: A Novel Image Segmentation Method for Rail Transportation	2023	The new segmentation model presents a hybrid of DeepLabV3 (semantic segmentation) and SAM (zero-shot segmentation). The resulting segments are highly precise and can be created quickly and at low-cost with very little sample data. When tested on RailSem19 dataset, DeepLabV3-SAM produced significantly better (up to 10%) mIoU results than DeepLabV3 or UNet; this is especially true for large objects with clear edges.
Road Extraction Based On Improved Deeplabv3 Plus In Remote Sensing Image	2022	This study presents a Way to improve Deeplabv3+, using a ResNeSt backbone and an ASPP block to provide the means for better road extraction from high resolution remote sensing aerial imagery. Improvements include improved continuity, and improved accuracy as well as improved mean Intersection over Union (IoU) compared to U-Net and traditional Deeplabv3+. There are still challenges for extracting roads with small widths or that are occluded, which leaves room for additional improvements in the future.

Segment Anything Model for Road Network Graph Extraction	2024	The method SAM-Road employs the SAM (Visual Transformer) encoder to capture visual information with the NMS (Non-Maximum Suppression) and GNN (Graph Neural Networks) from the transformer architecture to generate a representation of the road graph both accurately and quickly (40X faster than RNGDet++ without any kind of Post Processing). Unfortunately, it has difficulty detecting overpasses and elevated roadways and is limited because it only uses the SAM ViT-B encoder, which limits the overall accuracy of the segmentation.
Road-SAM: Adapting the Segment Anything Model to Road Extraction From Large Very-High-Resolution Optical Remote Sensing Images	2023	For road extraction, this paper presents the adaptation of SAM by incorporating three types of adapter configurations (i.e. series, parallel, mixed) combined with explicit visual prompting (EVP) to facilitate segmentation of the images. The proposed methods provide an accurate segmentation map from the segmentation models that have a lower computational cost to extract road features compared to other datasets, such as URUR or DeepGlobe. However, road extraction from complex urban scenes will require additional optimization.

Enhancing Road Scene Segmentation With an Optimized DeepLabV3	2024	This report outlines our efforts to improve the performance of the previously established DeepLabV3+ architecture for the purpose of detecting road scenes, including the use of a lightweight MobileNet V2 architecture instead of Xception as a backbone, improvements to the ASPP module and the addition of Shuffle Attention to increase the ability to perform real-time processing while maintaining excellent segmentation accuracy - specifically, for instances involving small objects as well as edges.
End-to-End Road Graph Extraction Based on Graph Neural Network	2023	The study introduces a method using Convolutional Neural Networks and Graph Neural Networks to obtain a roadmap. The approach detects major points of interest with CNNs and then predicts how they connect through Graph Neural Networks using Edge Convolution layers. This method does not use any post-processing steps, and it generally outperforms previous methods like Sat2Graph; however, because it relies on CNNs for finding major points of interest, it is susceptible to occlusions, which would influence the accuracy of predicted road connectivity.

Road Graph Extraction via Transformer and Topological Representation	2024	The Road2Graph model utilizes a transformer-based architecture and makes use of Deep Layer Aggregation to create road extraction systems from street topology represented as 7 dimensions. This allows the system to capture road interconnectivity accurately and produce accurate topological measurements (TOPO-F1, APLS) through topological metrics. However, the computational demand of this approach makes it impractical for real-time applications until it can be further optimized.
RNGDet++: Road Network Graph Detection Using Transformers	2023	An improved transformer-based model for detecting road networks through multi-scale feature fusion as well as instance segmentation has been developed. This new framework allows for more accurate detection of road shapes and complex intersection layout using a DETR-type architecture with imitation learning techniques. The performance achieved by this model is very accurate, but the computational load is very high and the inference time will be long, which restricts its use in systems that require real-time operation.

Single-Shot End-to-End Road Graph Extraction	2024	The current research paper suggests a totally convolutional neural network for identifying the road network by passing through only one stage. The totally convolutional neural network identifies the key points of the road network along with connectivity in just one go without the need to generate graphs iteratively. Notably, this approach has far better performance speed compared to previous networks while retaining high accuracy levels. However, this network fails in dense environments.

2.3 Software Requirement Specification

Requirements for the software and operating system to support the new road extraction framework.

The system is built to perform image segmentation using deep learning methods, hybrid segmentation, and fine-tuning using deep learning techniques based upon the Segment Anything Model (SAM).

2.3.1 Operating System

The supported platforms include the following:

- Windows 10 or newer

- Ubuntu 20.04 or newer

- Google Colab (Cloud system)

Google Colab is a great choice for training/experimenting with this system because it provides GPU support and is easy to set up.

2.3.2 Programming Environment

· Python version is at least 3.8

· Jupyter Notebook or Google Colab is recommended for use with this version of Python

· Visual Studio Code is an appropriate Code Editor/IDE to use for code development

Python was selected because it has a lot of built-in tools, that support Machine Learning as well as image processing.

2.3.3 Libraries and Frameworks

The software packages that are included in the implementation of the program consist of:

- The torch library is used for deep learning model creation, loading, and training with the SAM and DeepLabV3+ models.
- The segment-anything library is the official library utilized to implement the SAM.
- Torchvision provides access to many pre-trained models such as DeepLabV3+ that can be used in conjunction with utility functions for image transformations.
- Numpy is the primary library that handles the mathematical computations and handles the image arrays and masks in a more efficient manner.
- The opencv-python package (cv2) is used for image preprocessing, filtering, and edge detection.
- Matplotlib is used for visualizing and performing interactive point (prompt) selection.
- PIL (Python Imaging Library) is used for loading images and keeping them in a compatible format.

2.3.4 Model Dependencies

The system utilizes the following pre trained models:

SAM (Segment Anything Model):

- Pre trained weights such as ViT-B
- Obtained from official Meta AI's repository
- Used to perform the prompt-based segmentation as well as to fine-tune

DeepLabV3+:

- Pre trained model that uses ResNet backbone (in torchvision)
- Used for semantic segmentation and to generate coarse road masks

2.3.5 Hardware Requirements

A sufficient amount of processing power and memory will be necessary to complete training quickly. The specifications of what is required are as follows:

- Intel i5 processor or newer
- 8GB or more (recommended 16GB)
- NVIDIA GPU (recommended for training purposes unless otherwise specified) [for example RTX]
- At least 20GB free hard disk space for datasets and model files

Cloud based systems such as Google colab and other high-performance systems with cloud based Gpus would offer much more power to allow for much quicker completion of the trials or training.

2.3.6 Dataset Requirements

Used Dataset: DeepGlobe Road Extraction Dataset [22]

Input: High Resolution Satellite RGB Images

Annotations Include: Binary Masks (i.e., Road vs. Not Road)

The dataset will be utilized for training, validation and evaluation purposes in reference to segmentation models.

2.3.7 System Constraints

- Model training/fine-tuning requires a large amount of computational resources.
- The quality of the images used to train/generate the model and the quality of the prompts have an impact on the overall performance of the model.
- The availability of GPUs affects both how long it takes to train a model as well as how efficiently a model can be trained.

CHAPTER-3

Proposed Methodology

3.1 Design Methodology (UML representation)

3.1.1. Dataset Collection and Preparation

3.1.1.1Source and Input Description

High-resolution RGB satellite images and their associated binary road masks are obtained from the DeepGlobe Road Extraction dataset [22]. The sequential picture/mask pairs are used as inputs for supervised learning/model fine tune. The dataset contains diverse array of road types in both urban and rural settings, thus providing robust training for the model.

3.1.1.2 Quality Filtering and Cleaning

Corrupted images, noisy samples, and anyone who does not provide a visible road structure are eliminated from the dataset through thorough evaluation. By ensuring only useful, high-quality data are remaining in the dataset, model reliability is increased, and training's noise is decreased.

3.1.1.3 Dataset Structuring and Splitting

Typically, the training set consists of 80% of images and the validation set consists of 20%. The cleaned data (images and masks) will have the same dimensions (e.g., 512×512, 1024×1024) so that all data can be used the same way in the training pipeline.

3.1.1.4 Normalization and Mask Preparation

To standardise pixel distribution, input images are normalised using Equation (1):

$$I_{\text{norm}} = \frac{I-\mu}{\sigma} \tag{1}$$

where μ and σ represent mean and standard deviation of pixel values.

Masks are generated from the ground truth and converted to binary representation for road or background classification (binary pixel value = {0,1}). Efficient dataset loaders are utilised to optimise data access during training.

3.1.2. Data Preprocessing and Image Enhancement

3.1.2.1 Pixel Normalization

Normalization minimizes differences in sensor characteristics, and variations in lighting allow stable feature extraction and quicker training convergence.

3.1.2.2 Contrast Enhancement

Contrast stretching is applied using Equation (2) for the purpose of improving the visibility of very light or thin surface roads:

$$I_{\text{enh}} = \frac{I - I_{\text{low}}}{I_{\text{high}} - I_{\text{low}}} \times 255 \tag{2}$$

where I_{low} and I_{high} are chosen percentile intensities. The road textures that are hidden under shadows, vegetation, or low light become visible. The road edges are also enhanced, allowing them to be distinguished from the surrounding area.

3.1.2.3 Edge and Feature Enhancement

Using additional preprocessing techniques such as high-pass filtering or edge detection can help enhance road edges, especially where there is less contrast.

3.1.3. Adaptive Prompt Generation for Segmentation

3.1.3.1 Prompt Extraction from Ground Truth Masks

Rather than relying on manual prompting, adaptive prompts from ground truth masks are produced by the system. All ground-truth road pixels are positive and defined as road pixels, and background pixels are defined as negative, resulting in the identification of all background pixels.

3.1.3.2 Sampling Strategy

Random sampling is done to extract: - positive prompts from road sections - negative prompts from non-road sections This will help provide equal support to the segmentation model from both positive and negative.

3.1.3.3 Spatial Distribution of Prompts

Prompts will be spread out in the image to cover large road structures and any thin or fragmented areas, which improves both segmentation accuracy and consistency.

3.1.4. Segment Anything Model (SAM) Segmentation

3.1.4.1 Image and Prompt Encoding

The SAM model correctly generates image embeddings based on the input image's preprocessed attributes. The initial output also provides an example of how prompt points can be converted into prompt embeddings.

3.1.4.2 Mask Prediction

Using the Mask Decoder, the Segmentation Output is produced using Equation (3) via the encoded features from the encoder:

$$M_r = f_{\text{dec}}(E_I, E_P), P_{\text{refined}} = \sigma(M_r) \tag{3}$$

Where E_i represents image embeddings and E_P represents prompt embeddings.

3.1.4.3 Initial Segmentation Output

The segmentation masks output by the model identify all of the road region features (i.e., visual features) for the given image using both visual features and prompt (input) to generate the segmentation masks. The resulting initial mask outputs will need to be fine-tuned for domain type-specific accuracy.

3.1.5. Fine-Tuning of the Segmentation Model

3.1.5.1 Model Adaptation Strategy

Fine-tuning is performed on the pre-trained model with a domain-specific dataset to enhance its performance in a particular domain. The image encoder remains unchanged, whereas the prompt encoder and the mask decoder are trained.

3.1.5.2 Training Process

Training includes:

- Providing the model with inputs of images together with corresponding prompts
- Comparing predicted masks to actual ground truth
- Calculating loss and minimising through backpropagation.

3.1.5.3 Performance Improvement

Through Fine-tuning, the model learns road-specific features such as:

- Thin road structures;
- Connection patterns
- Details of boundaries.

This greatly enhances the quality of segmentation when compared to the results produced by a generalised model.

3.1.6. Post-Processing and Output Refinement

3.1.6.1 Morphological Operations

Morphological techniques to close (join) and open (separate) objects in space include filling (patching) small holes along road surfaces and eliminating lone (disconnected) noise pixels from an image.

3.1.6.2 Connectivity Enhancement

Furthermore, smoothing or using selective dilation will improve the continuity of roads while enhancing the connectivity of segments of a road.

3.1.6.3 Final Mask Generation

The segmentation results are saved as the final version of the road masks, which represent roads as precise and linear structures.

3.1.7. Evaluation and Result Analysis

3.1.7.1 Performance Metrics

The model's efficiency will be analyzed through the use of the standard evaluation metrics listed above:

- Intersection over union calculated using Equation (4)
- Dice coefficient calculated using Equation (5)
- Accuracy computed using Equation (6)

$$\text{IoU} = \frac{|M_{\text{pred}} \cap M_{\text{gt}}|}{|M_{\text{pred}} \cup M_{\text{gt}}|} \quad (4)$$

$$\text{Dice} = \frac{2|M_{\text{pred}} \cap M_{\text{gt}}|}{|M_{\text{pred}}| + |M_{\text{gt}}|} \quad (5)$$

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (6)$$

Where:

M_{pred} is Mask predicted

M_{gt} is ground truth mask

3.1.7.2 Visualization and Comparison

The resultant segmentation will be visualized together with the original satellite image, the ground truth mask and the predicted mask. This will enable a qualitative approach to analysing the final segmentation results, and improvements will be noted for segmentation accuracy from one source of information to another.

3.2 System Architecture Diagram

The goal of the architecture is to accelerate the extraction of roads from satellite imagery by leveraging features like: pre-processing, adaptive prompt generation, image segmentation using the Segment Anything Model (SAM) and post-processing of output. The way in which the architecture has been designed, is to incrementally increase the accuracy of the detected roads from its original satellite image to a connected road mask.

The architecture diagram is depicted in the Figure:3.1, the detailed description of each step is explained below.

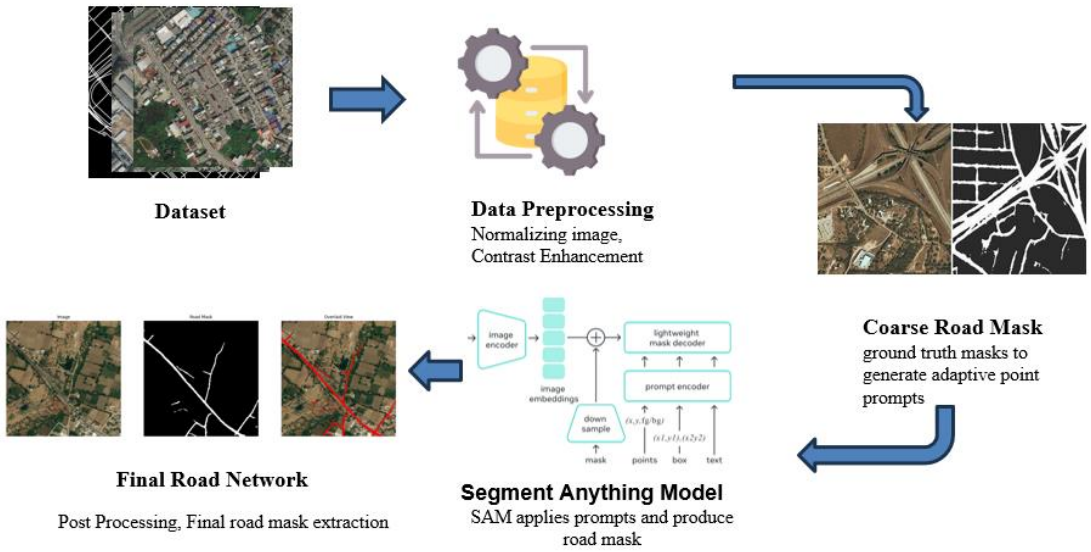


Figure 3.1: Architecture Diagram

3.2.1. Dataset Input

Working with the satellite images in the dataset [22] is how the process of generation begins. The images will have varying complexity for road structures due to differences in lighting, occlusion, and background noise.

•Source of input data: RGB images of the satellite

•Source of associated data: Ground truth masks of the roads associated with the arc data

Thus the generated outputs will be based off the input images.

3.2.2. Data Preprocessing

Prior to inputting the images into the model, there is a process known as "image preprocessing" that makes adjustments to improve the quality of the input images and bring them into a uniform standard across all images. Important image preprocessing techniques include:

- Normalization - The standardization of pixel intensity/ colour for a given image
- Contrast Enhancement - Improves visibility of very low colour/ contrast roads.

This stage of image preprocessing supports delineation of roads - distinguishing between very fine or faint roadways, particularly within low contrast areas.

3.2.3 Coarse Road Mask Generation

The dataset uses the ground truth masks to produce an understanding of where the road regions are located.

- These masks will allow for initial coarse representation of the road structures.
- They will provide a base from which adaptive point prompts can be defined.

This step represents a bridge between using more traditional knowledge of the type of road segmentation with new prompt learning methods.

3.2.4. Adaptive Prompt Generation

Point-based prompts are generated from the system by combining the coarse road masks and the following prompt types:

- Positive prompts are sampled from road areas of coarse regions;
- Negative prompts are sampled from property-type coarse regions.

These prompts serve as some of the spatial regions that will be identified and segmented by the model. The distribution of the prompts plays a role in assuring proper coverage of both the larger roadways and the smaller road segments.

3.2.5. Segment Anything Model (SAM)

The basic segmentation process is done by using the Segment Anything Model.

This model contains the following main parts:

- Image Encoder – This extracts image embeddings for the original image that is being processed in this case the satellite image
- Prompt Encoder - The spatial structure of the prompt will also be encoded
- Mask Decoder - This will combine both the image and prompt encoding to create the segmentation mask.

Visual characteristics and prompt suggestions are combined by SAM to produce better segmentation results for the road.

3.2.6. Post-Processing and Refinement

The output will be updated to achieve improved quality and connectivity after this phase is complete.

Improvements will be conducted on the following criteria:

- Elimination of unnecessary noise and small debris

- Increased connectivity of highways

- More consistent edge finishing for highway

The result of this phase will produce an accurately connected and clearly defined highway system representing the existing highway system in the real world.

3.2.7. Final Road Network Output

The finished product is a refined road mask illustrating the extracted road network. Overall, the improved road masks now better depict thin and occluded roads; show improved connectivity between the road segments; have drastically reduced false positives. As such, they can be used for analytical, visualization, and real-world applications such as mapping and navigation.

3.3 Description of Algorithms

3.3.1 Segment Anything Model Algorithm

Input:

- Satellite image I
- Ground truth mask M_{gt}

Output: Final refined road segmentation mask M_{final}

Step 1: Image Preprocessing

The satellite image will be resampled, normalised and enhanced using Equation (7) to highlight and delineate features:

$$I_{enh} = \frac{I - I_{low}}{I_{high} - I_{low}} \times 255 \qquad (7)$$

where:

- μ,σ→ mean and standard deviation
- $I_{low} I_{high}$→ intensity percentiles

Road features will be enhanced whilst noise will be reduced.

Step 2: Adaptive Prompt Generation

Prompts for model training will be automatically generated using the Equation (8) from ground truth mask data:

- P^{+}samples will be taken from road (R) pixel data
- P^{-} samples will be taken from background (B) pixel data

$$P = P^{+} \cup P^{-} \tag{8}$$

The generated prompts will give the model spatial constraints on how to find R pixels.

Step 3: Image Encoding (SAM Encoder)

An encoder (SAM) will process the enhanced satellite image using Equation (9) :

$$E_I = f_{enc}(I_{enh}) \tag{9}$$

where:

- $E_I \rightarrow$ image embeddings

Step 4: Prompt Encoding

The produced prompts are subsequently encoded as embeddings using Equation (10) :

$$E_P = f_{prompt}(P) \tag{10}$$

where:

- $E_P \rightarrow$ prompt embeddings

Step 5: Mask Prediction (SAM Decoder)

The embeddings will then be passed to a mask decoder using Equation (11) and used to produce segmentation outputs using Equation (12) for each record in the input image:

$$M_r = f_{dec}(E_I, E_P) \tag{11}$$

$$P_{mask} = \sigma(M_r) \tag{12}$$

Step 6: Fine-Tuning Optimization

The model will be fine-tuned / re-trained using ground truth mask data by minimising loss functions and the loss is calculated using the Equation (13) :

$$L = Loss(P_{mask}, M_{gt}) \tag{13}$$

The adjusted/neural parameters will update using the Equation (14) , via backpropagation, the parameters of the encoder and decoder objects in the model:

$$\theta = \theta - \eta \nabla L \tag{14}$$

where:

- $\theta \rightarrow$ trainable parameters
- $\eta \rightarrow$ learning rate

Step 7: Mask Binarization

The produced probability map will then be converted to a binary mask of either >0 or 0 using the Equation (15) :

$$M_{pred} = \begin{cases} 1, & P_{mask} >= 0.5 \\ 0, & otherwise \end{cases} \tag{15}$$

Step 8: Post-Processing

The produced binary mask will be improved via a combination of morphological operations to fill gaps in a mask; remove noise; and improve connectivity:

$$M_{final} = f_{post}\left(M_{pred}\right) \tag{16}$$

Step 9: Output Generation

The final image mask M_{final} is obtained using Equation (16) and is an example of the extracted road network, where continuity and accuracy have improved greatly.

3.3.2. Canny Edge Filter

To create a clear outline of a road, edge filtering, known as the Canny edge detector, will be performed on the satellite image and some more filteration techniques are also there which are given in Table 3.3. This will highlight both the boundary and linear paths characteristic of the road networks. The result will be a binary edge map that can be used to accurately place point prompts on the roads and place background prompts at a distance from the roads. This will improve the performance of SAM's ability to segment the road structures efficiently.

Steps involved:

1. **Original Grayscale Image**

• Convert the image to grayscale to allow edge detection to be performed on Intensity versus Color.

2. **Gaussian Blur (Noise Reduction)**

   - Reduce image noise by applying a Gaussian filter to the image (using Equation (17) .
   - **Formula:**

$$G(x,y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{17}$$

   Where:

   - σ=standard deviation of the Gaussian
   - *G (x, y)* = Gaussian kernel

3. **Gradient Calculation (Edge Strength and Direction)**

   - Compute the gradients in the x and y direction by using the Sobel operator.
   - **Formulas:**

- Horizontal gradient is computed using Equation (18):

$$G_x = \begin{matrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{matrix} \tag{18}$$

- Vertical gradient is computed using Equation (19):

$$G_y = \begin{matrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{matrix} \tag{19}$$

- Gradient Magnitude is computed using Equation (20):

$$G = \sqrt{G_x^2 + G_y^2} \tag{20}$$

- Gradient Direction is computed using Equation (21):

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right) \tag{21}$$

4. Non-Maximum Suppression

- Suppress pixels that are not local maxima in either gradient direction. This step will thin the edges. There is no direct equivalent method in OpenCV, it is instead processed when make a call to cv2.Canny().

5. Double Thresholding

Use both a low and a high threshold to classify pixels as follows:

- Strong Edges (above the high threshold)
- Weak Edges (below the high threshold and above the low threshold)
- Non-Edges (below the low threshold)

The purpose of this step is to prevent noise from being classified as an edge.

6. **Edge Tracking by Hysteresis**

- Trace the weak edges which are connected to any strong edge. If they are not connected then they will be discarded. Again, this is done in the cv2.Canny().

Table 3.3: Steps of Canny Edge Filtering

Step	Process	Purpose
1	Grayscale	Remove color complexity
2	Gaussian Blur	Smooth noise
3	Sobel Gradient	Find strength + direction of change
4	Non-Maximum Suppression	Thin edges
5	Double Threshold	Separate strong, weak, and non-edges
6	Hysteresis	Connect weak edges that belong

3.3.3 DeepLabV3 Algorithm

Input: Satellite/Aerial Image
Output: Road Segmentation Mask

DeepLabV3 is an architecture for semantic segmentation with a dual objective of capturing fine detail as well as larger context. To achieve its goals in segmenting roadways, it is adept at recognizing thin and linear structures while maintaining global coherence.

Step 1. Preprocessing

We first perform a normalization of the input image I by normalizing the intensity of pixels within the image to a common level. Normalizing helps to create uniformity between images such that processes within the network give rise to consistently developed features and stable models when trained using multiple input images over time.

Step 2. Feature Extraction Using Atrous (Dilated) Convolution using Equation (22)

The implementation of a ResNet backbone is executed for deep learning within the network by replacing conventional convolutional techniques in the deeper layers with atrous methods, which preserve for the class of input images being processed, the spatial resolution, while extending the receptive fields of the filters:

Main Formula – Atrous Convolution:

$$y[i] = \sum_{k=1}^{K} \quad x[i + r \cdot k] \cdot w[k] \tag{22}$$

where:

- r= dilation rate
- x= input feature map
- w= convolution kernel
- K= kernel size

In addition, this step permits long-range detail to be represented within the representation of roadways, particularly in locations where roadways may only have faint indications; therefore, detecting roads that connect over extended distances requires the ability to recognize long-range detail representation.

Step 3. Atrous Spatial Pyramid Pooling (ASPP)

The ASPP module utilizes several parallel atrous convolutions that comprise dilations of differing magnitudes within the network. This feature of the ASPP module aids in developing an understanding of the characteristics of roadway systems and supports the necessity of being able to identify the full range of roadway characteristics in images from narrow lanes to wide freeways.

The ASPP module consists of:

- 1×1 convolution

- 3×3 atrous convolutions with dilation rates r_1, r_2, r_3
- Image-level feature (global pooling)

All features are then concatenated to form a multi-scale context representation using Equation (23):

$$F_{ASPP} = \text{Concatenate}(F_1, F_2, F_3, F_4, F_5) \tag{23}$$

By generating a representation of multi-scale features, there is increased flexibility for the model to extract roadway data within complex backgrounds.

Step 4. Feature Fusion (Encoder–Decoder Architecture) using Equation (24)

Features originating from backbone low-level convolutional neural networks retain common elements of the environment like edge detection and boundary detection on roads, need to be synthesized with ASPP output after going through a process of changing dimensions.

$$F_{fused} = \text{Concatenate}(F_{up}, F_{low}) \tag{24}$$

Where:

- F_{up} is the upsampled ASPP output
- F_{low} contains high-resolution spatial details

Road Networks are thin structures and require precision to retain their borders when combining low-level feature maps with ASPP feature.

Step 5. Decoder and Upsampling

Once the features are fused, they will go through a refine process involving the convolution layers and an upsampling stage and it is done using Equation (25) :

$$F_{out} = \text{Upsample}(F_{dec}, H, W) \tag{25}$$

Step 6. Pixel-wise Classification (Mask Prediction) using Equation (26)

With the use of the softmax at every pixel location (x, y), the model is able to predict a class of either road or background from each pixel to provide the final mask $\widehat{M}$, which is a road segregation of the entire input image.

$$\widehat{M}(x, y) = \arg\max_{c} \; p(c \mid f(x, y)) \tag{26}$$

Where:

- $f(x, y)$= feature vector at pixel
- $c \in \{0,1\}$for background and road

3.4 Description of Datasets and Tools

The systematic use of benchmark datasets combined with powerful deep learning tools will result in the production of sustainable model that have been trained and validated to a high standard of quality. See the next section for a complete list of the datasets, frameworks and computer resources used in this phase of the work.

3.4.1. Datasets

DeepGlobe Road Extraction Dataset

In this study, we are leveraging the DeepGlobe dataset [22] to build an integrated SAM + DLv3 road extraction system. Due to its widespread use in remote sensing and image analysis for the purpose of road segmentation, it's recognized as a standard benchmark dataset.

Specs for the DeepGlobe Dataset include:

- High-resolution satellite images (50 cm to 1 m spatial resolution)
- All images have a 1024 pixel by 1024-pixel dimension
- The nature of the data is comprised of aerial images taken from places such as urban, rural, and semi-urban

- The ground truth of the data is provided as an annotated binary mask, where each pixel corresponds to a road or otherwise
- Diversity of road types in the data, including corners, intersections, narrow country roads, wide highways, and occluding features
- Types of environmental conditions encountered in the data include shadow, vegetation, different lighting, and cluttered environments.

The DeepGlobe dataset will allow for the evaluation of the effectiveness of the integrated SAM (prompt segmentation) and DLv3 (semantic segmentation) algorithms across multiple types of landscapes and in the field.

3.4.2. Tools

1. The SAM Pipeline, DLv3 Model Code, and preprocessing scripts will be created largely using Visual Studio Code (VS Code), an IDE developed by Microsoft and primarily used for web development.
2. The core programming language for building the SAM pipeline is Python (3.8+) which will allow you to integrate the SAM prompt generation module with the DeepLabV3 image segmentation model.
3. Google Colab (or "Colab"), is a cloud-based execution environment provided by Google, which is most commonly used for executing training, inference, and refinement of your model scripts. Colab provides access to powerful computing resources (GPUs) as well as the ability to easily manage files and have a simple way to test the integration of DLv3 and SAM without needing to set up your own hardware.
4. The pre-trained sampler weight files were (ViTS) were taken from Meta AI's repository. Model backbone (ResNet-50 and ResNet-101) pre-trained weights for DeepLabV3 (CLIP) will be retrieved from torchvision to extract road mask polygons. The utilization of pre-trained weights greatly minimizes training time as well as enhances performance on actual use cases of satellite imagery.

5. Hardware / Run time environment

- Development is on Windows 10+ using VS Code as the development platform
- Tensorflow used during the developing/testing of the neural network, as well as Google Colab to accelerate processing of models/outputs via GPU
- At least 10 GB needed on local disk for data (dataset/model/processed output)

CHAPTER-4

Results and Observations

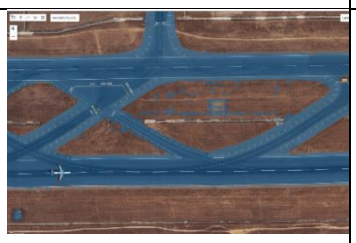
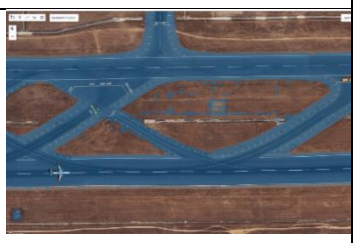
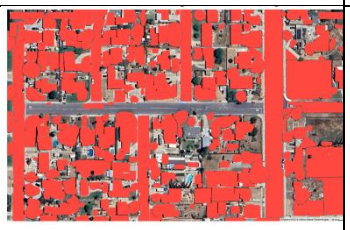
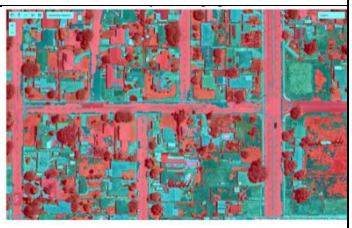
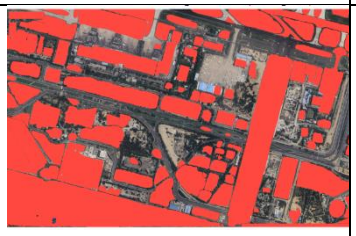
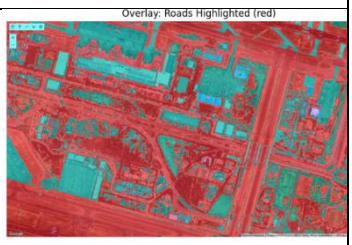
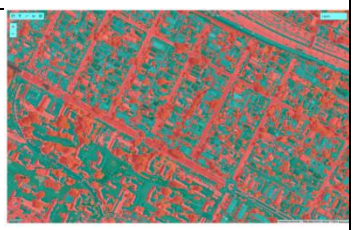
4.1 Step-wise Description of Results

The results achieved through different steps/points of the presented research work are detailed as part of this section/section results. The experiments were conducted using different methods which included the following: SAM-based segmentation; enhancements to prompt techniques; hybrid segmentation; and finally, using the fine-tuned sampling technique. The results have been summarized in the following ways based on the different types of satellite images: Rural; semi-urban; industrial and urban (dense). Each result has been provided with an input image; predicted mask; and an overlaid image showing the predicted mask on top of the input image.

4.1.1 Results of SAM-Based Segmentation with Edge-Aware Enhancement

Further information related to the limitations of using SAM in an all-automatic mode demonstrated through the qualitative results in Table 4.1 has been discussed above. For example, due to dense urban development, zero-shot segmentation will capture roofs, parking areas, as well as structures surrounding road surfaces resulting in over-segmentation and decreased precision of road boundaries. Similarly, with respect to narrow and partially obscured roads in industrial settings and/or residential layout, there is a strong probability that the model could lack the capability to maintain continuity (longitudinal) of the road and may either collapse, or merge, roads with surrounding features. With the use of edge-aware prompts and filtering methods, the segmentation outputs were more structured with distinct boundaries of road surfaces, reduced background noise and improved connectivity through intersection(s) and into narrow lanes. These improvements indicate that even though the original SAM model offers a solid general purpose segmentation base-line, enhancements specific to an area of application, i.e., road extraction, must be made to ensure consistent reliable performance for road extractions from high-resolution aerial and satellite images.

Table 4.1: Results obtained by SAM only and Sam with Filters

Original Image	**SAM only**	**SAM with filters**
		
		
		
		

Case 1: Wide Roads and Highways

In photos of broad highways with identifiable road structures asa shown in first row of table 4.1, SAM provides well segmented output with a strong, and continuous, segmentation. When identifying and capturing key-region segments in the predicted mask (the highway), minor non-highway areas were occasionally captured as noise. The results can be made to look cleaner, sharper, and with fewer false positive occurrences, through the use of edge-aware filtering techniques. That said, the resulting output is cleaner as

indicated by the improved overlay images when aligned to real world roads and exhibit a greater level of consistency and refinement.

Case 2: Suburban Residential Areas

According to the second row of Table 4.1, it is possible to identify large, well-defined highways in suburban areas clearly with SAM; however, narrow internal streets are frequently misrepresented as building edges or fragmented (i.e., broken) into segments due to having a small A width as well as being located in such complex surroundings. The edge-aware method was able to rectify these inaccuracies by reconnecting the broken segments, preserving the original road boundaries and providing increased distinguishability of the roads. Thus, the predicted masks exhibit greater continuity of road lines; and the overlayed road network illustrates improved structural integrity and alignment of the entire road network overall.

Case 3: Industrial and Complex Regions

In the industrial zones, in the third row of Table 4.1, SAM. will usually produce redundant or incorrectly classified outputs based on the fact that these industrial areas have large flat surfaces (large parking lots, large open land, etc.) and have road-like objects that are not accurately identified as road-like materials. Using edge-aware filters can significantly reduce the number of false positives by only filtering out non-road areas, while still keeping long linear shaped road-like objects. As you can see in the overlay, the process shows improved separation between the roads and other types of surface area that appear to be similar to roads and as a result will significantly improve overall segmentation accuracy.

Case 4: Dense Urban Environments

As depicted in last row of Table 4.1 ,SAM's outputs show fragmentation and significant gaps in the presence of thin roadways, while a few of the roads (e.g., road junctions) were incorrectly identified (e.g., showing a building structure versus a thin roadway). The modified method provides additional connectivity and reduces the amount of noise, which improves the visibility of thin roadways. Although there are still a few fractures in the modified overlay, the modified overlay's segmentation quality is noticeably improved over the SAM-only output.

4.1.3 Results of DeepLabV3 + SAM Hybrid Framework

The hybrid approach merges semantic segmentation and prompt-based refinement into one method.

Figure 4.1: Qualitative results of the proposed DeepLabV3 + SAM framework on a semi-urban region.

Figure 4.2: Qualitative results on a dense urban environment with complex structures.

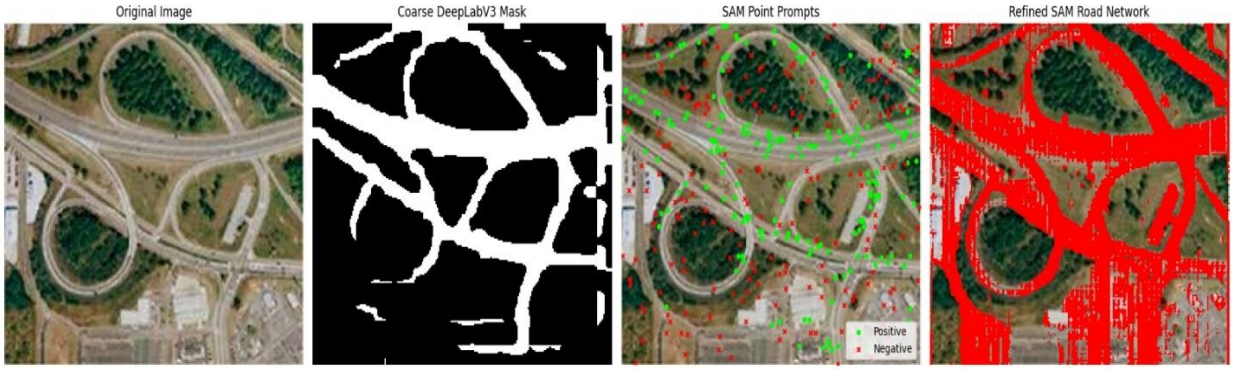


Figure 4.3: Qualitative results on a highway interchange with complex geometry.

Case 1: Structured Road Layouts

As depicted in Figure 4.1, The proposed mask that depicts the two different levels of roadway networks in urban areas has a similar portrayal to that of the identified roadway networks in the urban areas under consideration (as exhibited by the segmentation containing greater continuity and less gap relative to the actual roadway networks).

The findings from the overlaid studies show that there is an increased degree of agreement between the predicted and actual roadway networks.

Case 2: Complex Urban Scenes

Compared to the older techniques, this new approach can detect a greater number of roads. As shown in Figure 4.2, The model consistently detects narrower roads better than older models, but there is still noise in areas of high density. The model provides greater connectivity and a larger percentage of the road network in the overlay.

Case 3: Curved Road Networks

As shown in Figure 4.3, Predicted masks accurately replicate the underlying geometric structures of highway interchanges and/or curved road systems. Road geometry including curvature and intersection geometry are all captured accurately as a function of predicted masks. With regards to the overlay displayed above, it is evident that the overall structural consistency has improved.

4.1.4 Results of Proposed Fine-Tuned SAM Framework

The completion of the study validates the SAM-based segmentation framework that we proposed previously in multiple environments, including rural, semi-urban and dense urban areas. Findings show that the model is capable of generalising across different degrees of complexity, while maintaining the continuity of roads and structural accuracy.

Figure 4.4: Qualitative results on a semi urban area

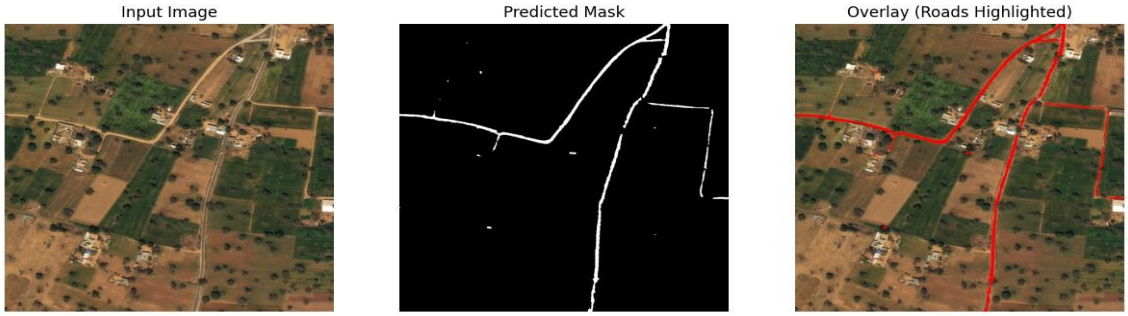


Figure 4.5: Qualitative results on a rural area road network

Figure 4.6: Qualitative results on a semi urban region with complex road network

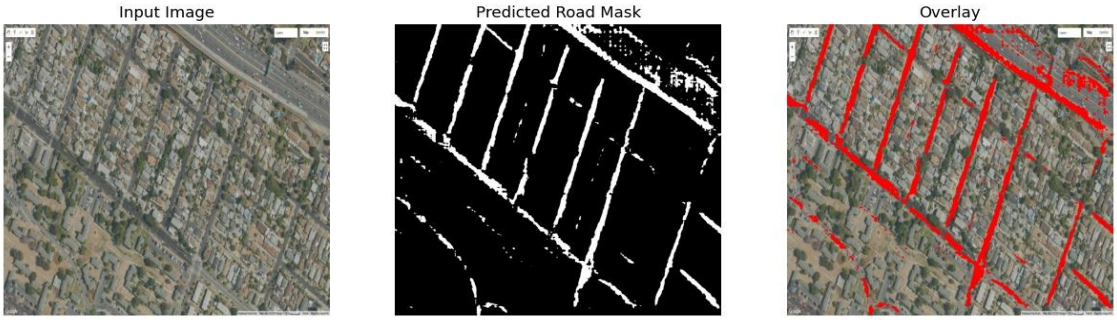


Figure 4.7: Qualitative results on a dense urban grid structure

Case 1: Semi-Urban Road Network

The conceptual model outlines an urban-area case with a freeway, a series of local highways and roads, and surrounding green vegetation as depicted in figure 4.4.

The predicted image shows the main freeway as a continuous limb extending across the entire freeway system, which is connected to a series of narrow local highways and roads that are aligned in a structured manner. The segmented road boundary has improved precision and exhibited less noise than previous techniques.

The overlay of the segmentation and aerial image clearly illustrates a high level of agreement between the predicted and real highway systems; thus demonstrating an effective means of distinguishing highways from nearby non-highway areas (e.g., buildings, parks).

Case 2: Rural Road Network

The input image contains a rural scene consisting mainly of houses surrounded by empty fields and light-coloured, poorly defined gravel roads as shown in Figure 4.5.

The predicted manner of the mask displays the largest rural thoroughfares, whilst also displaying the smaller paths that link the fields to the homes. The segmentation output shows good connection from the roadways to the adjacent surrounding terrain with very few disturbances in the background, indicating that the roads are reasonably and consistently segmented from the surrounding area.

The overlay of the segmentation output with the real roadway positions shows very good correlation between the real and predicted roadway position while preserving the long continuous characteristics of the roadways, without introducing any inaccurate detections.

Case 3: Semi-Urban Road Network (Moderate Complexity)

This type of research project involves a semi-urban environment consisting of large streets, residential areas, and surrounding buildings and trees as shown in Figure 4.6.

The developed mask has detected major arterial highways as well as secondary internal roads with good accuracy. Road intersections and patterns of division have been

represented well from a connection standpoint, however, there may be some minor errors in areas that are heavily congested with cluttered objects.

The overlays show good correspondence with the actual road networks (connectedness) while retaining their respective properties throughout various segments of the road network and depicting an overall road structure throughout the research property.

Case 4: Dense Urban Grid Structure

In this case as shown in figure 4.7, an urban area contains dense urban buildings that are placed closely together and have a grid type of road system, including many very narrow streets.

The result of the prediction shows the grid pattern roads being correctly predicted and includes the roads having some parallel streets and intersections. The majority of the road segments were accurately detected with very little fragmentation; however there may be some minor fragmentation due to high density or occlusion.

The overlay visualisations show that the overall structure of the network (i.e. road topology) has been well preserved, clearly showing the structured layout and maintaining connectivity throughout the network.

4.2 Result Analysis and Test Case Results

Different methods for segmentation of roads will be analyzed during this section:

1. SAM based segmentation (baseline and enhancement by edge-aware methodology)

2. Hybrid using SAM and DeepLabv3

3. Fine-tuned SAM segmentation

Each of the three methods will be assessed qualitatively and quantitatively through a variety of test cases representing rural, semi-urban and highly developed urban environments.

4.2.1 Qualitative Analysis

The visual output depicted in section 4.1 illustrate a clear improvement in the quality of segmentation across each of the three methods. The SAM - only configuration shows a reasonable ability to generalise in simple environments (i.e. open roadways and highways), however its difficulty to detect narrow roads, fragmented output and high amounts of false positive (incorrect classifications of a segment) occur in complex environments such as heavily populated urban or industrial regions.

Although the use of an edge-aware algorithmic enhancement process to improve the continuity of roadways and the definition and connectivity of their boundaries was demonstrated to improve these parameters, the model still has no semantic understanding of visually similar types of structures within its segments and has produced significant amounts of incorrect classifications of visually similar structures as well.

The implementation of a hybrid DeepLabV3 + SAM approach has also exhibited a significant increase in the performance level of this segmentation approach due to the incorporation of semantic (contextual) information with the use of segmentation masks that have been produced by the DeepLabV3 will facilitate the provision of additional structured placeholders for SAM resulting in an improved level of road continuity and an improvement in the definition and connectivity of the boundaries of segmented items while reducing the number of false positive classifications from the original SAM Only method. In addition, this new hybrid approach has displayed an ability to successfully perform in structured environments and moderately complex environments including curved roadways; however, it has increased computational complexity and increased reliance on multiple models to achieve its successful performance.

According to literature review, the proposed SAM Framework is the most effective of the three frameworks to conduct mapping from this research. The fine-tuned SAM model uses domain specific training datasets to adapt the model parameters. Adaptive prompt generation allows for the creation of adaptive prompts that will adequately capture wide and narrow roadways and roads in all environments. The analysis demonstrated this by displaying an increased level continuity and a higher level of road topology preservation while also achieving a significant reduction in the number of false positive detections. The

proposed framework improves performance relative to the hybrid method used previously, and does not rely on a segmentation model; therefore, the proposed method is both more efficient and more scalable.

4.2.2 Quantitative Analysis

Standard segmentation metrics, like Intersection over Union (IoU), F1-score, Precision, and Recall will be used as further indicators of performance. The performance displayed in the below table 4.2.

Table 4.2: Quantitative Analysis

Method	**IoU (%)**	**F1-Score (%)**	**Precision (%)**	**Recall (%)**
SAM (Minimal Prompting)	54.3	70.4	68.1	73.0
SAM (Edge-Aware)	60.7	75.6	72.8	78.7
DeepLabV3 + SAM	75.8	86.2	84.7	87.9
Proposed Fine-Tuned SAM	**82.4**	**90.1**	**88.6**	**91.3**

4.2.3 Interpretation of Results

Evaluation of the quantitative results shows:

- The performance of the SAM baseline is adequate, but lacks the level of precision needed for advanced tasks.
- The edge-aware enhancement primarily helps to improve performance through better resolution of boundary locations, which in turn helps to improve recall for detected elements within a scene. However, very limited increases in performance were noted by the edge-aware method due to limited semantic knowledge of features being detected.
- There are significant improvements (approx. 15-20%) in IoU between the SAM + DeepLabV3 Hybrid models versus SAM only models. This demonstrates the value of using semantic information.

- The present fine-tuned SAS model performs best across all elements tested, and thus produces more accurate results from 6-8% ICN relative to the Hybrid models.

The improvements in precision demonstrate that there are fewer false-positive detections, while the increase in recall shows more accurate detections of thin and occluded roadways. The increase in the F1-score illustrates that both precision and recall were balanced through the enhancement of each.

4.2.4 Test Case Analysis

Comparison of the performance in each environment is shown below:

- Rural Areas: Although all methods provide good results, the proposed method shows much better ability to detect faint or narrow roads.
- Semi-urban Areas: The hybrid and proposed methods are superior to those using only SAM methods as they have better overall connectivity and have better detection of road structures within a developed area.
- Dense Urban Areas: In dense urban areas, the proposed method performs the best at detecting road networks while maintaining an accurate representation of grid patterns with fewer false positives, but challenges still exist caused by a high amount of visual clutter.

4.2.5 Summary

The experimental findings show beyond a doubt that the suggested Fine-Tuned SAM Framework outperforms other frameworks in the three critical areas of Accuracy, Connectivity, and Robustness regardless of the environment. Additionally, this framework provides high quality segmentation while still being efficient and therefore practical for use in real-world applications such as extracting roads from images.

4.3 Observations from the Work

The evaluation through experiments of the different segmentation methods shows that there is a clear evolutionary trend from baseline to the proposed fine-tuned framework; while the SAM (Segment Anything Model) has good generalization capabilities for use in simple

environmental tasks, it experiences difficulties when detecting thin roadways, producing fragmented output and misclassifying within complex environments. Edge-aware prompting improves the precision and continuity of road boundaries, but it does not offer any semantic understanding.

The hybrid DeepLabV3 approach provides additional structural guidance than either of the baseline models, resulting in improved continuity of the road and a reduction in false positives, though at the expense of increased complexity within the system. The proposed fine-tuned SAM framework demonstrates superior performance based on the learning of domain-related roadway features; as a result, it correctly identifies the presence of wide and thin roads in diverse environments.

Observably, there are large increases in road continuity, boundary precision and decreased false positive errors as a result of using the proposed fine-tuned SAM framework compared to either of the baseline models. The results of the study indicate that the use of the fine-tuned model, along with adapting prompting, significantly increases the segmentation performance and thus will provide benefit for real-world applications associated with roadway extraction.

CHAPTER-5

Conclusion and Future Study

5.1 Conclusion

The focus of this research was on enhancing the performance of road segmentation from satellite images using the Segment Anything Model (SAM) and its modified versions. The study explored various methodologies in order to determine which methods would improve road segmentation. The various methodologies tested were the baseline SAM segmentation method, the edge-aware prompting method, hybrid segmentation (DeepLabV3) combined with the SAM method, and the proposed fine-tuned SAM framework.

The results of the experiments show that while the SAM model has good performance with less complex scenarios, the model struggles to detect thin roads and has difficulty in more complex environments. Using the edge-aware techniques helped to improve the precision and connectivity of the edges, however, there was still a lack of understanding of the meaning of the data at these locations. The hybrid DeepLabV3 + SAM model improved upon this issue by using semantic information to aid in the segmentation of the images, thus improving the overall quality of the segmentation.

The results of the proposed fine-tuned SAM model were the best results obtained for all of the methods evaluated. The fine-tuning of the SAM model allowed it to adapt to the specific characteristics of the area; therefore, the model was able to successfully identify the roads within rural, semi-urban and urbanized areas. The results of the testing showed that the fine-tuned SAM model provided the best overall performance in regard to road continuity, road edge accuracy, and reduction of false positive details, and do not require the use of separate modelling approaches.

In summary, this research indicates that using a general-purpose segmentation model and using specific knowledge of the specific area to fine-tune the model will provide significant improvements in the performance of the model for real-world applications of extracting roads.

5.2 Future Work

The new design has demonstrated high quality results thus far. However, the design still has not reached its full potential and it has many areas where improvements can be made. The system probably needs to support extensions and be more adaptable to assist with handling complex urban situations (for example, finding roads through heavy occlusion (i.e. a significant obstruction of the road by other physical objects) or shadowing (i.e. the darkness created by the obstruction of the sun by other objects) through providing additional contextual data from the nearby area and/or multiple modalities of data (i.e. elevation or models of captured images and when they were captured) to increase the quality of the segmentation process).

Future work will continue to explore ways to further enhance the ability to detect very thin and low-contrast road features by incorporating advanced enhancement techniques. Additionally, there is potential to develop the ability to process in Real-time as well as on less-formidable and/or resource-constricted systems.

Lastly, creating a road network extraction (as graph structures) in the framework will enable increased navigation and traffic analysis capabilities. Future efforts will also focus on finding ways to apply transformer-type architectures with advanced methodologies for prompt generation to increase the robustness and overall effectiveness.

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Appendix

```
#SAM inference Function
def predict(image, sam, device):
    sam.eval()
    image = cv2.resize(image, (1024, 1024))
    image_tensor = transforms.ToTensor()(image).unsqueeze(0).to(device)
    with torch.no_grad():
        image_embedding = sam.image_encoder(image_tensor)
        # Same prompt as training
        points = torch.tensor([[[512, 512]]], device=device)
        labels = torch.tensor([[1]], device=device)
        sparse_embeddings, dense_embeddings = sam.prompt_encoder(
            points=(points, labels),
            boxes=None,
            masks=None
        )
        low_res_masks, _ = sam.mask_decoder(
            image_embeddings=image_embedding,
            image_pe=sam.prompt_encoder.get_dense_pe(),
            sparse_prompt_embeddings=sparse_embeddings,
            dense_prompt_embeddings=dense_embeddings,
            multimask_output=False,
        )
 mask = torch.nn.functional.interpolate(
            low_res_masks,
            size=(1024, 1024),
            mode="bilinear",
            align_corners=False
        )
        mask = torch.sigmoid(mask).cpu().numpy()[0][0]
        mask = (mask > 0.5).astype(np.uint8)
    return mask
```

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  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
5% Publications
6% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.